Institution's Innovation Council
MHRD's Innovation Cell, AICTE

Innovation / Prototype
(Product/Service/Process)

Team Details	
	Team Lead:

Name	Email	Contact no.
Sweta Mohan	221001902@rajalashmi.edu.in	8838430630

Team Members Details:

Sr. No.	Name	Email	Contact no.
1.	Seethalakshmi S	221001149@rajalashmi.edu.in	9042993291

Add more fields if required

Mentor (if any)

S.No	Name	Email	Contact no.
1.	Dr.K.E.Narayana	narayana.ke@rajalakshmi.edu.in	9884454498

Add more fields if required

Institute Name and Address: RAJALAKSHMI ENGINEERING COLLEGE (AUTONOMOUS), CHENNAI-602 105

Title (20 Words Max)	Early Stage Bell Pepper Leaf Disease Detection Using Deep Learning-Based Capsule Network with Convolutional Neural Network
***Developed as part of**	Academic Requirement/Study Project
***Choose the Financial Year, during the Idea-PoC/Innovation Developed**	2025-26
***Sector / Domain**	Agriculture & Rural Development
Innovation Type	☐ Product

INSTITUTION'S INNOVATION COUNCIL
(Ministry of HRD Initiative)

***Development Stage - Technology Maturity of the Solution/Innovation in terms of Technology Readiness Level TRL (if applicable (Refer TRL Stages)**	TRL 6 – Prototype system tested in intended environment (web app + real images)
Development Stage - Manufacturing Maturity of the Solution/Innovation in terms of Manufacturing Readiness Level (MRL) Refer MRL Stages	MRL 4 – Capability to produce in laboratory environment
Development Stage: Investment Readiness Level of the Solution/Innovation (IRL). Refer IRL Stages	IRL 4 – Low-fidelity MVP (working prototype system developed)

***Define the problem and its relevance to today's market / society / industry need (Max: 100 Words)**	Bell pepper crops are highly vulnerable to foliar diseases, leading to significant yield losses and economic impact on farmers. Traditional disease detection relies on manual inspection, which is time-consuming, subjective, and often fails to identify infections at an early stage. Existing AI models also struggle with real-world challenges such as image distortion, lighting variation, and lack of actionable recommendations. There is a critical need for an automated, accurate, and intelligent system that can detect diseases early, assess severity, and provide practical treatment solutions to improve agricultural productivity and sustainability.
***Describe the Solution / Proposed / Developed (Max: 100 Words)**	The proposed solution is an AI-powered system that detects bell pepper leaf diseases using a hybrid CNN-CapsNet architecture. The system processes leaf images through preprocessing, segmentation, and feature extraction to identify disease presence and severity. It achieves high accuracy in classification and provides severity-based treatment recommendations. The solution is implemented as a web-based application, enabling farmers to upload images and receive instant diagnosis along with actionable insights. Additionally, it includes a follow-up monitoring feature, transforming it into a complete decision-support system for precision agriculture.
***Explain the uniqueness and distinctive features of the (product / process / service) solution (Max: 100 Words)**	The uniqueness of the solution lies in its hybrid CNN-CapsNet architecture, which combines efficient feature extraction with advanced spatial understanding. Unlike traditional models, it preserves part-whole relationships, making it robust to image distortions such as rotation and noise. The system not only detects diseases but also quantifies severity using lesion percentage and provides tailored treatment recommendations. Integration of follow-up monitoring further enhances its functionality. This combination of accuracy, robustness, and actionable intelligence makes the solution highly effective for real-world agricultural applications.
***How your proposed / developed (product / process / service) solution is different from similar kind of product by the competitors if any (Max: 100 Words)**	Unlike conventional CNN-based models that struggle with spatial distortions and provide only classification outputs, this solution integrates Capsule Networks to enhance spatial awareness and robustness. It goes beyond simple detection by incorporating severity estimation and actionable treatment recommendations. Additionally, the system includes a continuous monitoring feature, enabling dynamic tracking of disease progression. Compared to existing tools, which often lack practical decision support, this solution offers a comprehensive end-to-end framework, making it more suitable for real-world deployment in agriculture.
Is there any IP or Patentable Component associated with the Solution?	No

Has the Solution Received any Innovation Grant/Seed fund Support?	No
Are there any Recognitions (National/International) Obtained by the Solution?	No
Is the Solution Commercialized either through Technology Transfer or Enterprise Development/Startup?	No
Had the Solution Received any Pre-Incubation/Incubation Support?	No
***Utility: Highlight the utility/value proposition (key benefits) aspects of the solution/innovation* (Max: 100 Words)**	The system provides early disease detection, severity assessment, and actionable treatment recommendations, helping farmers take timely decisions. It reduces crop loss, improves yield quality, and minimizes excessive pesticide usage. The platform acts as a digital agronomist, making expert knowledge accessible to farmers. Its web-based interface ensures ease of use and accessibility, even in rural areas. Overall, it enhances productivity, reduces costs, and supports sustainable farming practices.
***Scalability: Highlight the market potential aspects of the Solution/Innovation (Potential Market Size, segmentation and Target users/customers etc.) (Max: 100 Words)**	The solution has high scalability potential as it can be extended to multiple crops and diseases by training with additional datasets. It can be deployed as a mobile or cloud-based application, making it accessible to farmers globally. The target users include small-scale farmers, agricultural experts, and agribusiness companies. With integration into IoT and smart farming systems, it can enable real-time monitoring at scale, making it suitable for large agricultural markets.
***Economic Sustainability: Highlight commercialisation/business application aspects of the solution (how it is going to economic profitable and viable) (Max: 100 Words)**	The solution can be monetized through subscription-based services, pay-per-use models, or integration with agricultural platforms. It reduces financial losses for farmers by preventing crop damage and optimizing resource usage. Agritech companies and government agencies can adopt it for large-scale deployment. With increasing demand for smart farming solutions, the system has strong commercial potential and long-term financial viability.
***Environmental Sustainability: Highlight environmental friendliness aspects and related benefit of the solution/innovation (Max: 100 Words)**	The system promotes eco-friendly agriculture by reducing excessive pesticide usage through precise disease detection and severity-based recommendations. It minimizes chemical runoff, protects soil health, and reduces environmental pollution. By enabling early intervention, it prevents large-scale crop damage and supports sustainable farming practices. Overall, it contributes to environmental conservation and aligns with green agriculture initiatives.

Video URL	https://drive.google.com/file/d/1kKZmDbXcVc0ouqRpZ9VMAJzR7CEFYhTf/view?usp=drive_link
Upload Photograph: (JPG, PNG, PDF max 2 MB)	https://drive.google.com/file/d/1fyvecUxMhk3LbyDC0Qb6o7f0hYt-n0x8/view?usp=drive_link

(e) Timeline- Develop a timeline against the milestones for taking Idea/PoC to Prototype Development and (or) Commercial level/start-up stage.	Idea & Research – 1 month Data Collection & Model Design – 2 months Model Development & Testing – 2 months Web App Development – 1 month Prototype Completion – 1 month Future: Commercialization – 6–12 months
Applicability of Solution 10 Marks *(Max 50 words for each from a-e)*	
(a) **Usability:** what is the usability of your innovation. *(Level of acceptance of innovation and its Features among target group)*	The system is user-friendly and accessible via web/mobile platforms. Farmers can easily upload images and receive instant results, making it highly usable even for non-technical users.
(b) **Scalability:** how your innovation will be scalable at market level.	The solution can scale across different crops, regions, and platforms, supporting large agricultural communities through cloud-based deployment.
(c) **Economic sustainability:** Explain the potential of innovation to become profitable or financially viable.	It reduces crop loss and input costs, providing financial benefits to farmers and offering monetization opportunities for developers.
(d) **Environment Sustainability:** How your innovation is environment friendly or address environmental problems.	Minimizes pesticide overuse and promotes sustainable farming, reducing environmental impact.
(e) Is there any Intellectual Property (IP) Component associated with innovation? if yes, explain.	Currently no patent, but the hybrid CNN-CapsNet model and decision-support integration have potential for future IP filing.
Define the potential market size (in terms of INR) and target customers. *(Max 100 words)*	The global precision agriculture market is rapidly growing and expected to reach over ₹1 lakh crore. In India, the agriculture technology market is expanding due to increasing adoption of AI-based solutions. Target customers include small and large-scale farmers, agricultural consultants, agri-tech companies, and government organizations. The solution can also be integrated into smart farming systems and mobile applications, making it highly relevant in both domestic and international markets.

INSTITUTION'S INNOVATION COUNCIL (Ministry of HRD Initiative)

Annexure 1

Themes:

1. Healthcare & Biomedical devices.
2. Agriculture & Rural Development.
3. Smart Vehicles/ Electric vehicle/ Electric vehicle motor and battery technology.
4. Food Processing.
5. Robotics and Drones.
6. Waste management.
7. Clean & Potable water.
8. Renewable and affordable Energy.
9. IoT based technologies (e.g. Security & Surveillance systems etc)
10. ICT, cyber physical systems, Block chain, Cognitive computing, Cloud computing, AI & ML.

10 stages of MRL

MRL 1: Basic manufacturing implications identified
MRL 2: Manufacturing concepts identified
MRL 3: Manufacturing proof of concept developed
MRL 4: Capability to produce the technology in a laboratory environment
MRL 5: Capability to produce prototype components in a production relevant environment
MRL 6: Capability to produce a prototype system or subsystem in a production relevant environment
MRL 7: Capability to produce systems, subsystems or components in a production representative environment.
MRL 8: Pilot line capability demonstrated. Ready to begin low rate production.
MRL 9: Low rate production demonstrated. Capability in place to begin Full Rate Production.
MRL 10: Full rate production demonstrated and lean production practices in place.

9 stages of TRL:

TRL 0 : Idea. Unproven concept, no testing has been performed.
TRL 1 : Basic research. Principles postulated observed but no experimental proof available.
TRL 2 : Technology formulation. Concept and application have been formulated.
TRL 3 : Applied research. First laboratory tests completed; proof of concept.
TRL 4 : Small scale prototype built in a laboratory environment ("ugly" prototype). TRL 5 : Large scale prototype tested in intended environment.
TRL 6 : Prototype system tested in intended environment close to expected performance. TRL 7 : Demonstration system operating in operational environment at pre-commercial scale. TRL 8 : First of a kind commercial system. Manufacturing issues solved.
TRL 9 : Full commercial application, technology available for consumers.

INSTITUTION'S INNOVATION COUNCIL (Ministry of HRD Initiative)

INSTITUTE READINESS LEVEL

IRL 1: Basic Research (Need Identification & Peer Review Publications) & Completed First-Pass Business Model Canvas (BMC)

IRL 2: Applied Research (Market Size and Competitive Analysis) & Business Plan – Value Proposition & IP Identification

IRL 3: Validate Problem - Solution Fit (Confirmed Value Proposition & Techno-Economic Analysis) & Minimum Product Cost (Maturity of Core Technology)

IRL 4: Prototype Low-Fidelity Minimum Viable Product (MVP): "Low-fidelity" - A representative of the component or system that has limited ability to provide anything but initial information about the end product.

IRL 5: Validate Product-Market Fit (Integrated Validation of the Minimum Viable Process and Process Engineering). "High-fidelity" - A high-fidelity laboratory environment would involve testing with equipment that can simulate and validate all system specifications within a laboratory setting.

IRL 6: Validate Business/Revenue Model: Integrated Pilot Development– understanding operational nuances

IRL 7: Prototype High Fidelity MVP: Integrated Pilot Continuous Operation

IRL 8: Pre-Commercial Demonstration – Operating Conditions and quality stabilized

IRL 9: Full Commercial Development – A full time process engineering staff continuously verifies that operations are meeting cost, yield and productivity targets.